A065-v0.4.1 | Arc A770 3D Euler numerical audit

Release category: NUMERICAL SCREENING ONLY

This report records device-executed FFT/RK4 tests only when an actual Intel Arc A770 Level Zero device is identified. Missing hardware produces NOT_RUN_PREREQUISITE, never PASS. Synthetic Gaussian-initialized Euler data do not establish physical PKLSA, Casimir, gravity, or independently observed pressure.

Pre-registered inputs

A003 source geometries	torus_3_1_direct / torus_3_1_hoste
Grid ladder	48^3 / 64^3 / 96^3
Numerical cube	L = 12 periodic
Initial vortex core	Gaussian a = 0.55 (assumed)
Circulation / density	Gamma = 1; rho = 1
Euler integrator	Fourier projected rotational RK4; 4 steps, T = 0.02
GPU	SYCL Level Zero / dpnp FFT FP32
Reference	NumPy FFT FP64, same initial data
Precision warning	Previous DD32 Biot-Savart parity is NOT FFT/Euler parity

New gates G43-G48

Gate	Result	Interpretation
G43	NOT_RUN_PREREQUISITE	MISSING_DPNP_DPCTL: install compatible Intel dpnp and dpctl into active venv
G44	NOT_RUN_PREREQUISITE	MISSING_DPNP_DPCTL: install compatible Intel dpnp and dpctl into active venv
G45	NOT_RUN_PREREQUISITE	MISSING_DPNP_DPCTL: install compatible Intel dpnp and dpctl into active venv
G46	NOT_RUN_PREREQUISITE	MISSING_DPNP_DPCTL: install compatible Intel dpnp and dpctl into active venv
G47	NOT_RUN_PREREQUISITE	MISSING_DPNP_DPCTL: install compatible Intel dpnp and dpctl into active venv
G48	UNRESOLVED	No independently calibrated PKLSA physical vortex core / heldout velocity and pressure

GPU unavailable: MISSING_DPNP_DPCTL: install compatible Intel dpnp and dpctl into active venv

Authority boundary

G48 remains UNRESOLVED. The Euler solver integrates assumed geometry-derived initial vorticity; no external physical transverse-vorticity, force or pressure holdout is supplied. Historical failed gates are unchanged.